ME2 — Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

Alyanna Lopez · October 2026 · two-head Conformer-S · 246,055 params · 16,628 clips · Trained on NVIDIA A100 (CUDA) · Validated on RPi 4/5

MODEL

Architecture	two-head Conformer-S (1 layer · 64-d · 4 heads)
Parameters	246,055
Weights	1.03 MB float ONNX · 0.32 MB int8
Input feature	log-mel(40) · 100 frames · 1 s
Outputs (dual)	intent_logits [B,20] + slot_logits [B,19]
Slotted intents	6 (timer, alarm, temp, bright, color, reminder)

DATASET

Source	HF · airimonda/ai231-me2-voice-commands
Clips / hours	16,628 clips · 9.6 h · 16 kHz mono
Speakers	307 train / 115 test / 5 holdout (disjoint)
Voice mix	synthetic + real (train)
Labels	20 intents · 18 slot values · 6 slotted
Split (speaker-disjoint)	10,733 / 4,443 / 1,250 neg / 202

TRAINING ON NVIDIA A100 (CUDA)

Cluster	NVIDIA A100 (CUDA) · 1 GPU · seed 0
Objective	joint CE: intent (20-way) + slot (19-way)
Optimizer	AdamW (lr 3e-4, wd 1e-4, LS 0.1) · seed 0
Epochs	95 (early-stop, best 80) · train loss 2.12
Schedule	cosine annealing (T_max 100) · batch 64
Weighting	real-voice ×5 · OOS ×3 · slot CE ×1
Wall-clock	≈7.2 min · 1 node
Encoder	1 Conformer layer · 64-d · 4 heads

VALIDATION (HELD-OUT, UNSEEN SPEAKERS)

Intent accuracy	85.6% [84.5, 86.6] · bal. 84.1%
Slot exact match	69.5% end-to-end · 77.2% given intent
Macro-F1 / OOS	0.839 macro-F1 · OOS FA 32.5% (106/326)
Real vs synthetic	44.2% real · 94.4% synthetic
Worst / best intents	OOS 60.9 · Vol↑ 73.0 / Remind 94.2 · Timer 94.1
In-scope precision @τ0.7	97.1% precision · 92.2% recall
Inference latency	p50 0.74 ms · p95 0.88 ms (M-series CPU)
Wake word (openWakeWord)	"hey mycroft" · peak 0.9996 · 0/250 FA @0.75
INT8 parity	99.2% argmax match · 0.32 MB (–69%)
Runtime	onnxruntime · int8 · CPU · fully offline

PIPELINE — MIC TO ACTUATOR



SUBMISSION READINESS

What the reviewer will look for

Status of each deliverable for this submission

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REVIEWER CHECKLIST

1	Repo public, one-command reproduction	done · launch_web.sh + setup_pi.sh
2	Dataset licensed and citable (DOI)	done · HF DOI 10.57967/hf/10723
3	Training logs + final checkpoint committed	done · best.pt + train_log.json (5 archs)
4	Pi 4/5 latency reproduced by the posted script	pending on-Pi re-run · p50 0.74 ms (M-series)
5	Held-out test set, unseen speakers	done · 115 test + 5 holdout, disjoint
6	Baseline of comparable size compared	done · CNN/MLP/TCN/CRNN vs Conformer-S

REFERENCES & ARTEFACTS

GitHub repository	github.com/alyannalopez/MLOps-Exercises
Dataset location	HF · airimonda/ai231-me2-voice-commands · DOI 10.57967/hf/10723
Compute	NVIDIA A100 (CUDA) · 1 node · 95 epochs (best 80) · seed 0
Model weights	conformer_s_float.onnx (1.03 MB) · conformer_s_int8.onnx (0.32 MB)
Wake word	openWakeWord "hey mycroft" · MIT · dscripka/openWakeWord

Winner of the 5-architecture benchmark: two-head Conformer-S (246k params), test accuracy 85.6% [84.5, 86.6] on 4,443 held-out clips from 115 unseen speakers; slot exact-match 69.5%. All figures taken from report/stage5_evaluation.md and report/stage6_deployment.md.